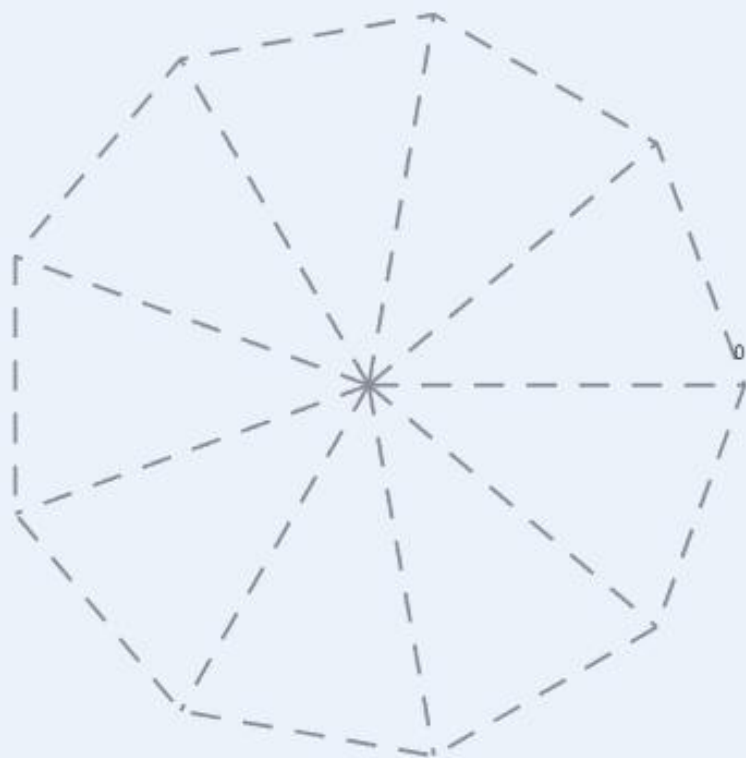




Choose at most three topics that you find most difficult (multiple vote on a single choice is allowed; max 3 votes)



- A 0% Find Cartesian from Parametric Equations
- B 0% Find Tangent Lines to Curve
- C 0% Implicitly defined parametrizations
- D 0% Area enclosed by curve
- E 0% Length of Curves
- F 0% Graph the set of polar coordinates points
- G 0% Polar to Cartesian Equations
- H 0% Cartesian to Polar Equations
- I 0% Nothing is difficult



Ex. 10.2 Q18

$$x \sin t + 2x = t$$

$$t \sin t - 2t = y$$

$$t = \pi$$

Aim: Slope of the curve $\left[\frac{dy}{dx} \right]_{t=\pi}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \dots = \text{function of } t, x(t), y(t)$$

$$\frac{dy}{dt} : t \sin t - 2t = y$$

$$\text{Diff. w.r.t } t \Rightarrow \underline{t \cos t + \sin t - 2} = \frac{dy}{dt}$$

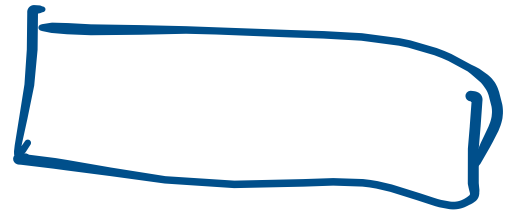
$$\frac{dx}{dt} : \underline{x \sin t} + 2 \underline{x} = t \quad \begin{array}{l} x \text{ is a func. of } t \\ \underline{x(t)} \end{array}$$

$$\Rightarrow \underline{\frac{dx}{dt} \sin t} + x \cos t + 2 \underline{\frac{dx}{dt}} = 1$$

$$\Rightarrow \frac{dx}{dt} = \underline{(1 - x \cos t)} / (\sin t + 2)$$

$$\frac{dy}{dx} = \frac{t \cos t + \sin t - 2}{\left(\frac{1 - x \cos t}{\sin t + 2} \right)}$$

sub. $t = \pi, x = \pi/2$



When $t = \pi$, $x \sin t + 2x = t$

$$x \sin \pi + 2x = \pi$$

$= 0$

$$2x = \pi \Rightarrow x = \pi/2$$

$$y = t \sin t - 2t \Rightarrow y = -2\pi \quad (\text{No need})$$

① Find $\frac{dy}{dt}$, $\frac{dx}{dt}$

② Find $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

③ Find values of x and y (set $t=a$)

Can just find those
appear in dy/dx

④ Find $\frac{dy}{dx} \Big|_{t=a}$.



We want to find the area under $y = x^3$ over $[0, 1]$

. Choose the correct expression for the area using the parametrization $x = t^2, y = t^6, 0 \leq t \leq 1$.

☐ $\int_0^1 2t^6 dt$

☐ $\int_0^1 2t^7 dt$

☐ $\int_0^1 6t^6 dt$

☐ $\int_0^1 6t^7 dt$

Responses Hidden



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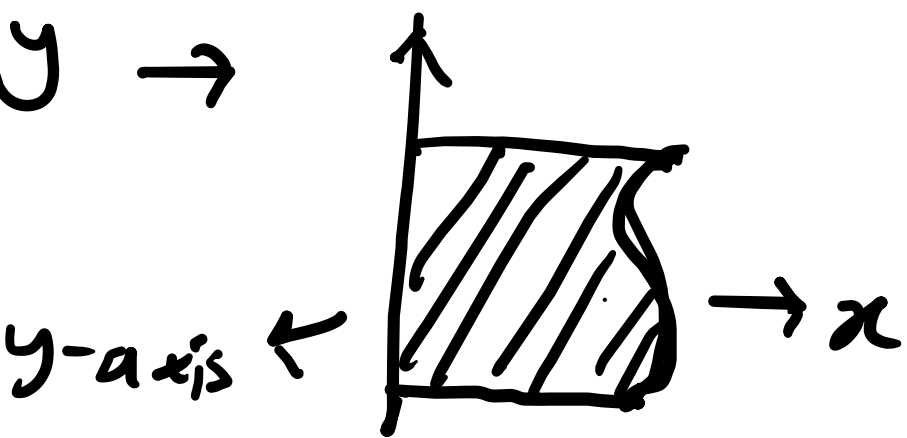


Types of regions:

① $\int y \, dx \rightarrow$



② $\int x \, dy \rightarrow$



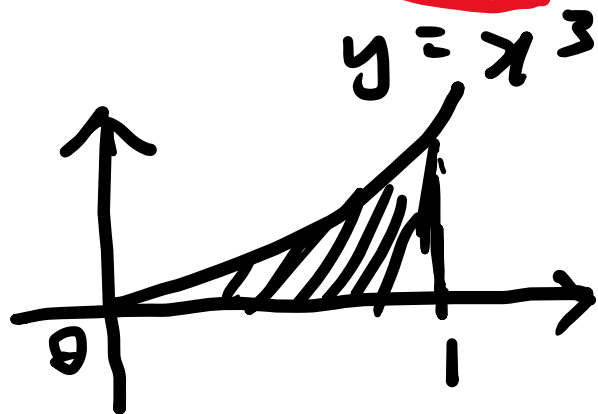
Ex 10.2 Q24

Area under $y = x^3$ over $[0, 1]$

(a) $x = t^2$, $y = t^6$, $0 \leq t \leq 1$ ✓

$$0 \leq t \leq 1$$

$$\Rightarrow 0 \leq x \leq 1$$



$$[-1 \leq t \leq 0]$$

↳ This ^{maybe} gives
negative area

$$\text{Area} = \int y \, dx$$

$$\text{Area} = \int_0^1 \underline{y} \, \boxed{dx}$$

$$= \int_0^1 t^6 \cdot 2t \, dt$$

$$= \int_0^1 2t^7 \, dt$$

$$= \dots$$

$$x = t^2$$

$$y = \underline{t^6}$$

$$x = 0 \Rightarrow t = 0$$

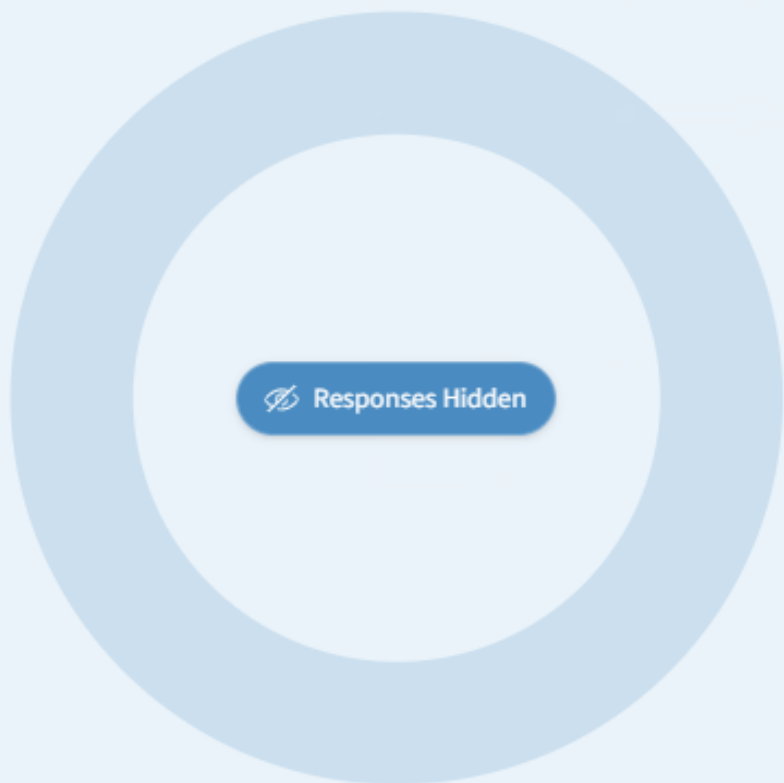
$$x = 1 \Rightarrow t = 1$$

[reject $t = -1$]

$$dx = 2t \, dt$$



Choose the correct graph for $0 \leq \theta \leq \pi/2$, $1 \leq |r| \leq 2$



A

0%



B

0%



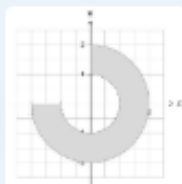
C

0%



D

0%



Method to graph:

- ① Look at θ
- ② Identify regions for $r > 0$, $r < 0$.
- ③ Look at r , find intersection.

Ex. 10.3 Q26

$$0 \leq \theta \leq \pi/2$$

$$1 \leq |r| \leq 2$$

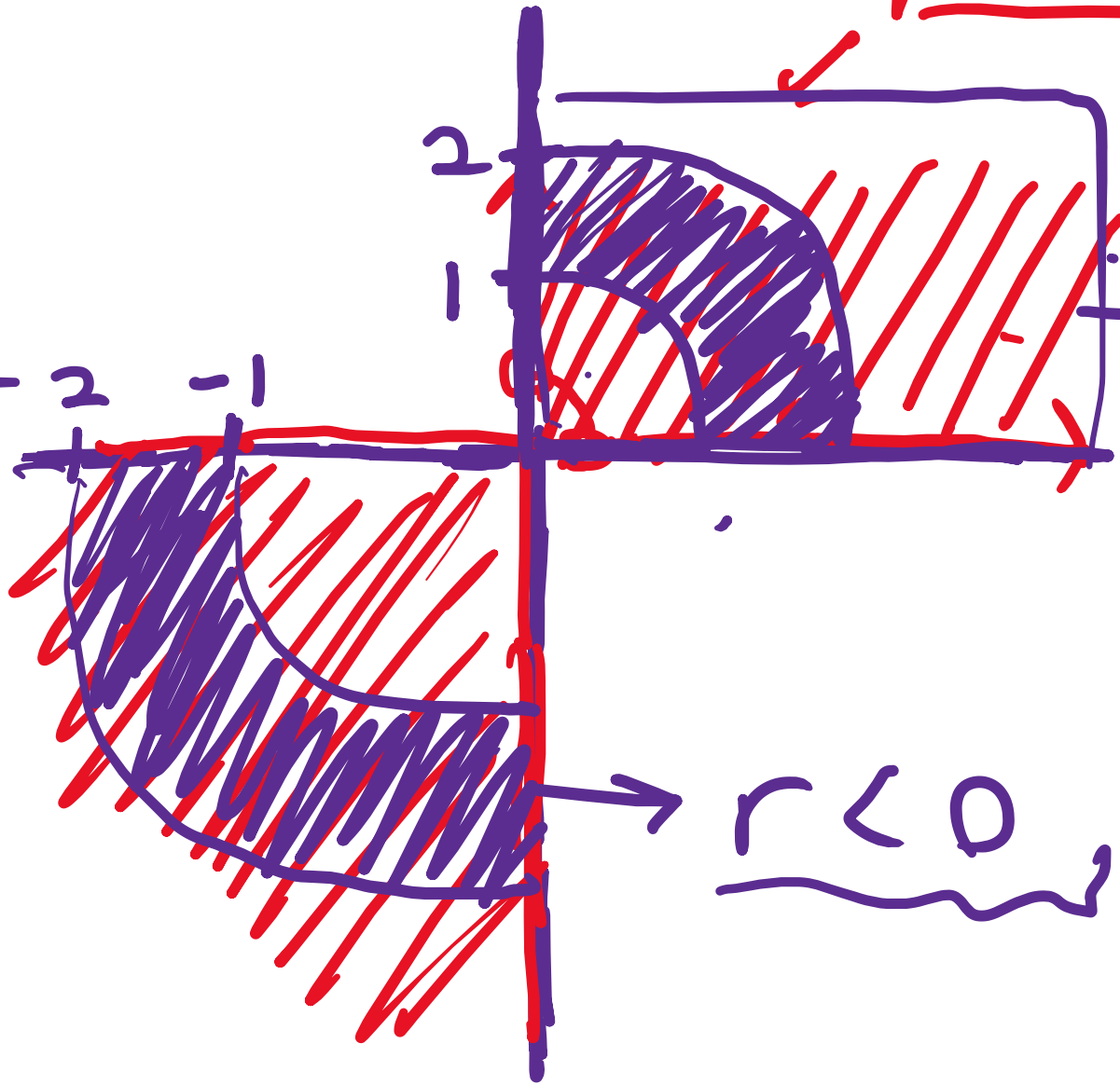


$$1 \leq r \leq 2$$

or

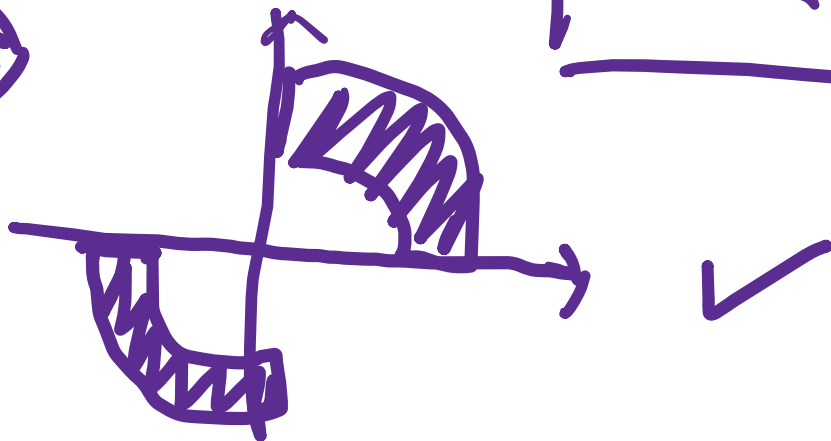
$$1 \leq -r \leq 2$$

$$-2 \leq r \leq -1$$



$$r > 0$$

$$r < 0$$

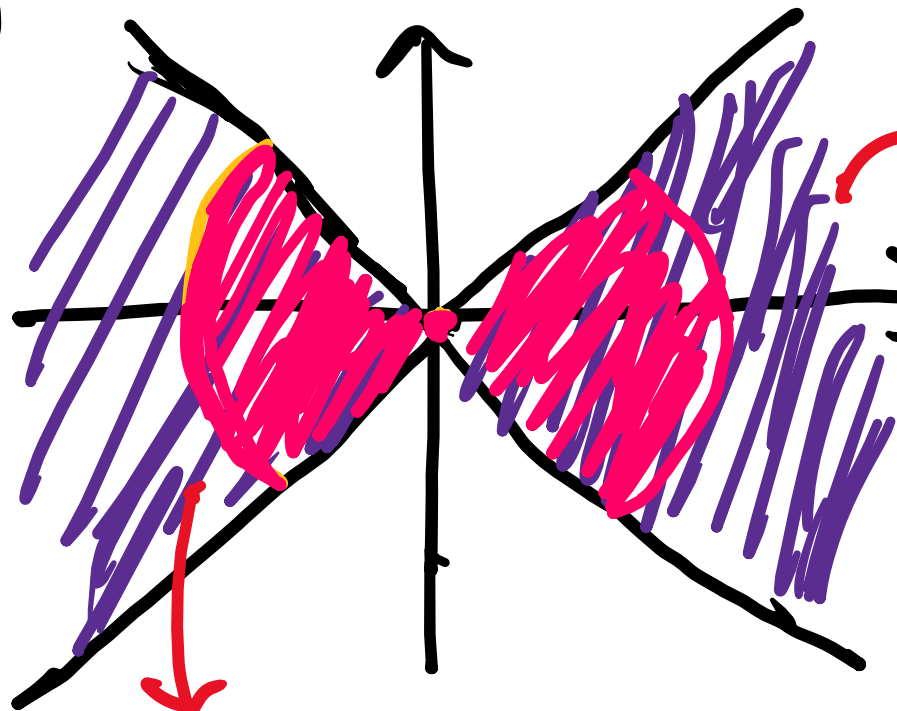


Ex 10.3 Q24

$$\underline{-\pi/4 \leq \theta \leq \pi/4}, \quad \underline{-1 \leq r \leq 1}$$

$$\boxed{-1 \leq r < 0}, \quad \boxed{0 \leq r \leq 1}$$

Q1

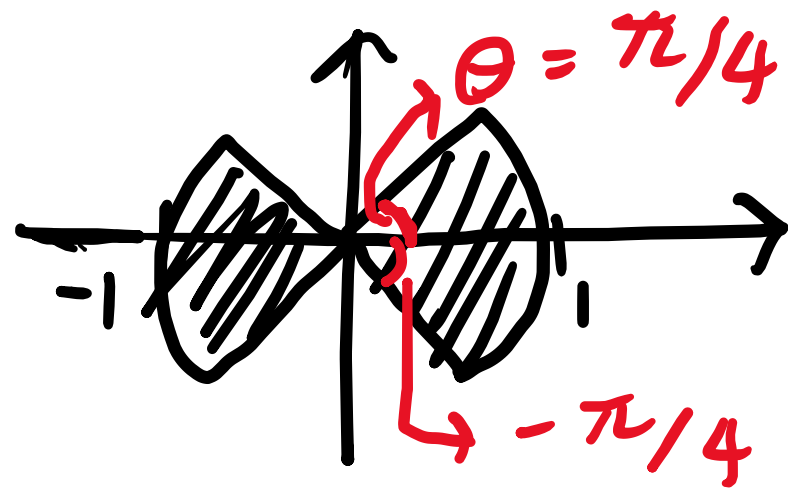


$r < 0$

$r > 0$

$\Rightarrow$

Answer:





When converting cartesian equation $(x - 5)^2 + y^2 = 25$ to polar equation, I write

$$x^2 - 10x + 25 + y^2 = 25 \implies x^2 + y^2 = 10x \implies r^2 = 10r \cos \theta \implies r = \pm \sqrt{10r \cos \theta}$$

. What do you think about this argument?

Sounds good to me.

It is flawed.

It is flawed but I cannot think of a reason for that.

It is technically correct but it gives no information on the values of

r .

Correct:
~~use~~ $r^2 = 10r \cos \theta$

$r = 10 \cos \theta$ (divided
by r)



Responses Hidden

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Show Correct Answers [A]